

Group Semester Projects

[Version 6.0: September 2026]

The goal of *Applied Bioinformatics* is to train students in practical analysis of complex biological data, using microbiome data as an entry point. Thus, the best way to assess these skills (computational competencies, quantitative analysis, and application of domain expertise) will be with a practical application. The majority of the grade (see course overview document) will be based on this group semester project.

Students will form small teams, ideally based on *shared interests* and *complementary skill sets* (e.g., from different programs/backgrounds), to solve a real problem using real biological data and a suite of bioinformatics skills learned in this course. Teams will analyze their data by applying data analysis and critical thinking skills learned during the course (see Specific Tasks at the end of this document). All students are expected to run the analysis workflow and contribute to all parts of the assessment, but balanced delegation of other tasks (e.g., different sections of the written report; different analytical sections) is allowed, provided that all students participate in all tasks, delegation is balanced and fair, and justification is provided in the work plan. If specific tasks are delegated, this should be described briefly in a contributions statement in the group report.

During the first few weeks of the course:

1. **Form a team of 3 students.** Projects will be presented by the instructors, and students will rank their favorite projects using a poll. Students are organized into groups based on shared interests. Final groups will be approved by the instructors.
2. **Project challenges** will be presented to each group of students. Each group will get a unique challenge, and a unique dataset! Start planning immediately to think over the problem as a group and brainstorm ways that the problem might be solved. *Do not worry if you do not have any clue right now*, just jot down your notes and revisit later in the semester. The goal is for groups to work on their project during each week of the semester (including class time) to apply each week's lessons to their project dataset. Solutions will become more apparent as the semester proceeds.

By the end of Module 2 (mid-term!):

1. Groups should have met on a weekly basis (minimum) to discuss the challenge and to start planning solutions.
2. **Midterm Report** (5% of overall project grade) is due (see timeline)
 - a. This is a brief (1 page) report giving:
 - i. A short summary of the problem *in your own words* (one paragraph).
 - ii. A short description of what you have done so far (one paragraph) and any early results or problems encountered.
 - iii. A short description and Gantt chart describing your plans for addressing the problem by the end of the semester, indicating which tasks are assigned to each group member (one paragraph plus chart)
 - b. What should have been done by now:
 - i. Data should be successfully imported and explored in a Jupyter notebook.

- ii. Sequence data should be demultiplexed, denoised (amplicon data), and quality filtered, as appropriate (i.e., apply procedures learned in the first weeks of this course). You should perform an initial analysis of taxonomic composition.
 - iii. Your code should be organized, concise, and well-annotated. Any notebooks should be fully executable (including data download and importing). These do not need to be submitted with the report, but you should develop best practices early.
- c. The grade will be assigned to the entire group, based on:
 - i. A demonstrated understanding of the problem (20%).
 - ii. A clear and systematic plan for how to address the problem (50%).
 - iii. A clear and feasible timeline and work plan (30%).
 - iv. Reports that simply repeat what is written in the project description will not receive full credit.
- 3. Groups are encouraged to meet during the exercise session with the teaching assistant to discuss their progress and raise any concerns.

At the end of the semester:

1. **Final written reports are due before the presentation sessions!** (see timeline/Moodle for due date)
2. Groups will present their projects in class, as an oral presentation with powerpoint slides (slides are also submitted on Moodle before the presentation date).
 - a. Presentations will last 10 minutes maximum (strict limit!), followed by Q+A (note that Q+A will focus on the presentation and report contents, but individual questions may relate to any course contents).
 - b. Groups should present:
 - i. An overview of their challenge, with a clear framing of the problem and the group's hypothesis or key goals (1 min suggested).
 - ii. An overview of their methodology (3 min suggested; high-level detail).
 - iii. Their main results/solution (5 min suggested)
 - iv. Key conclusions (1 min).
 - c. Each member of the group should take turns presenting.
3. Evaluation:
 - a. Group members are evaluated individually based on their presentation and answers to questions.
 - b. Grading criteria: clarity, correctness/scientific quality, time management (within time limit, appropriate balance of topics and group members), understanding of the methodology and problem, answers to questions.

Final Report:

The final report is the culmination of the semester work, and will comprise two components:

1. The analysis notebooks for the project, worth 50% of the total project grade.
2. A final report, in the form of a short research article (format below), worth 45% of the total project grade.

Part 1: Analysis code (Jupyter Notebooks)

Each project will organize their work using Jupyter Notebooks that they create on JupyterHub. Groups should perform all project steps (data analysis and documentation) in these notebooks. All final code and notebooks should be submitted on Moodle as a single directory/Zipped archive OR via a version-controlled repository (e.g., GitHub).

1. Submit all analysis code as Jupyter notebooks, separating distinct steps into separate notebooks (e.g., upstream quality control vs. different downstream analyses) following a clear and consistent naming scheme. All notebooks should be fully executable and well commented to describe the rationale for different steps; raw data can be retrieved via URL requests, and small data files can be submitted with the notebooks together via Moodle to make downstream analyses executable.
2. Establish a clear internal structure (e.g., separate data from analysis notebooks and any code modules in separate directories).
3. Give clear filenames to all files, and use titles/headers/text to clarify what these files are and if there is any logical progression to them. Remove any unnecessary files, e.g., intermediate outputs or preliminary notebooks that are not relevant for the final report.

The grade will be assigned to each individual based on (1) clarity of code (e.g., code comments, style), (2) appropriate use of methodology, (3) executability of code, and (4) equal contributions of all members.

Part 2: Written report

This should be written in the form of a short research article (maximum 5 printed pages single-spaced size 10 text, excluding figures and bibliography) and **submitted in PDF format**. Hence, it should be professional, concise, and clear. ***You must not use large language models (LLMs) for writing any parts of your report.***

The report itself will have a specific structure:

1. Abstract (≤ 200 words) describing the problem, methodology, results, and conclusion in 1-2 sentences each. Should clearly present problem and findings
2. Short introduction (~ 250 words). Briefly describe the problem, and provide background on the biological context (e.g., describe why this problem is important and cite primary literature that has examined similar problems).
3. Methods (~ 500 -1000 words). Concisely describe the methodology used to address the problem, and the rationale for this approach. This should cite the primary literature appropriately. Consider using a figure (e.g., flowchart) to give an overview of your methodology.
4. Results (~ 1500 words). Concisely describe your findings, interpretation, and solution. Use figures and tables to show your findings, and avoid overly verbose descriptions of these findings — instead, distill these findings to demonstrate a clear understanding of the results and their interpretation.
5. Discussion (~ 250 -500 words). Do not repeat the results. Instead, connect your findings and interpretation with the larger problem. Discuss limitations of your approach. Discuss future steps that could be taken to better address the problem. Describe your overall conclusion. Consider using another figure, if appropriate, to convey your conclusion.
6. References. Cite all primary literature, methods, etc, appropriately! Make sure that these are high-quality references in high-impact journals and well researched; your instructors know this literature well and will not be impressed if you are missing the key, landmark articles on the topic and just pulling the first reference you can find from a google search!
7. Additional figures, text, etc, can be placed in the appendix.

The grade will be based on:

1. Abstract + Introduction: Clear understanding of the problem and discussion of background (5%)
2. Methods + Results: Appropriate application of methodology and presentation of methods (25%)
3. Results + Discussion: Clear presentation of results and interpretation, demonstrating understanding of methods, approaches, and their limitations (45%)
4. Data visualization: Quality figures, interpretability, style (15%)
5. Overall quality and depth (10%).

All students are expected to contribute equally to the writing. Students who do not contribute equally to the writing and presentation may have points deducted.

The following basic rubric gives a guideline of grade levels:

Grade Range	Explanation
< 4	Failed to understand the fundamental theory and practical application of bioinformatics. Data and code are disorganized. Failed to link understanding of bioinformatics theoretical concepts to biological interpretations in their data. Elements of data and code are missing, disorganized, and/or not reproducible. Very poor coordination among team members.
4.0 – 5.0	Demonstrate adequate understanding of the theory, methods, and practical applications. Adequate analysis of biological data, yielding generic conclusions that do not exceed the minimum expectations. Notebooks, data, and code are adequately organized to the extent that the evaluators can assess how all steps were performed, and repeat the analysis if needed. Some errors, flaws, and misunderstandings. Team work was not well coordinated or balanced.
5.0 – 5.5	Demonstrate a high level of understanding of the theory, methods, and practical applications. Demonstrated high commitment with an ambitious and challenging project aim that went beyond the basic expectations, with insightful and well written interpretations of the data that show a clear understanding. Notebooks, data, and code are well organized with a clear and transparent internal structure, fully reproducible. Very few errors. Team work was very well coordinated and balanced.
5.5 – 6.0	Demonstrate a very high level of understanding of the theory, methods, and practical applications. Demonstrated very high commitment and creative thinking, going above and beyond the basic project expectations to plan and execute an ambitious and highly challenging project aim that yields novel insights into the data, with a clear understanding of the project challenge and finding inspiring solutions in bioinformatics. Notebooks, data, and code are extremely well organized with a clear and transparent internal structure, fully reproducible. Flawless! Team work was very well coordinated and balanced.

Specific Tasks:

Each group will receive an individual project assignment, with some specific questions that are unique to that group's dataset. In addition, all groups are expected to utilize all or most of the methods learned in this course (and maybe some others as well!), except when it is not possible or appropriate for their data (ask an instructor

if you think this is the case). Below are some specific tasks that should be performed or attempted by all groups. Each of these tasks should be accompanied by one or more visualizations or tables (as appropriate):

1. **Sequence quality control.** Appropriate quality control and filtering procedures should be applied. For amplicon datasets, appropriate denoising and/or clustering techniques should be applied, except if a group is given explicitly pre-processed data. For metagenome datasets, appropriate QC methods should be applied to assess assembly and binning quality.
2. **Taxonomy classification.** Appropriate taxonomy classification techniques should be applied and explained. Try different methods and databases — you do not need to include this in your report, but you could if the differences are meaningful (e.g., one database clearly gives better results).
3. **Alpha diversity** should be estimated for all samples, and compared across the primary sample categories* (e.g., sample types or other main groups) and/or gradients* (e.g., age, space, time, pH, or other continuous sample metadata). Test out a few different metrics (including kmer-based and/or phylogenetic metrics) to see what they reveal. Apply appropriate statistical tests.
4. **Beta diversity** should be measured and compared between the primary sample categories or gradients*. Test out a few different metrics (including kmer-based and/or phylogenetic metrics) to see what they reveal. Apply appropriate statistical tests.
5. **Which features are more/less abundant in different groups?** Apply appropriate statistical tests and/or supervised learning methods to answer this question. Consider carefully what types of features you want to compare between groups — maybe use different feature definitions (e.g., collapsed on taxonomy or functional information).
6. **Functional profiling or prediction.** For metagenome data, perform functional annotation. For 16S rRNA gene amplicon data, use q2-picrust2 to predict the metagenome composition of your samples. Use this to compare your primary categories or gradients*. Repeat beta diversity analysis based on predicted metagenome composition. If functional prediction is not possible on your dataset (e.g., you only have fungal ITS data), choose another substantive method as a replacement (see below).
7. **Do something new X2 (or X3).** Part of this course is learning how to read and use bioinformatics documentation! Pick substantial methods that we did not learn about in class, and which will add value to your group project (especially if it relates to one of the specific objectives of your group project). Install it, use it, and present your results! This could be a QIIME 2 plugin or action that we did not use, or different substantial statistical or visualization packages for Python. Choose at least two additional methods that are of high significance and substance; or three if you did not perform functional prediction. Simply using different databases/metrics with existing methods does not count, unless if you clearly demonstrate significant additional effort that yields an insightful new analysis (e.g., benchmarking many methods or databases side-by-side).

*The “primary categories or gradients” of interest in your project will be described in your assignment sheet, but don’t stop there! Other exciting things could be found in the sample metadata provided.